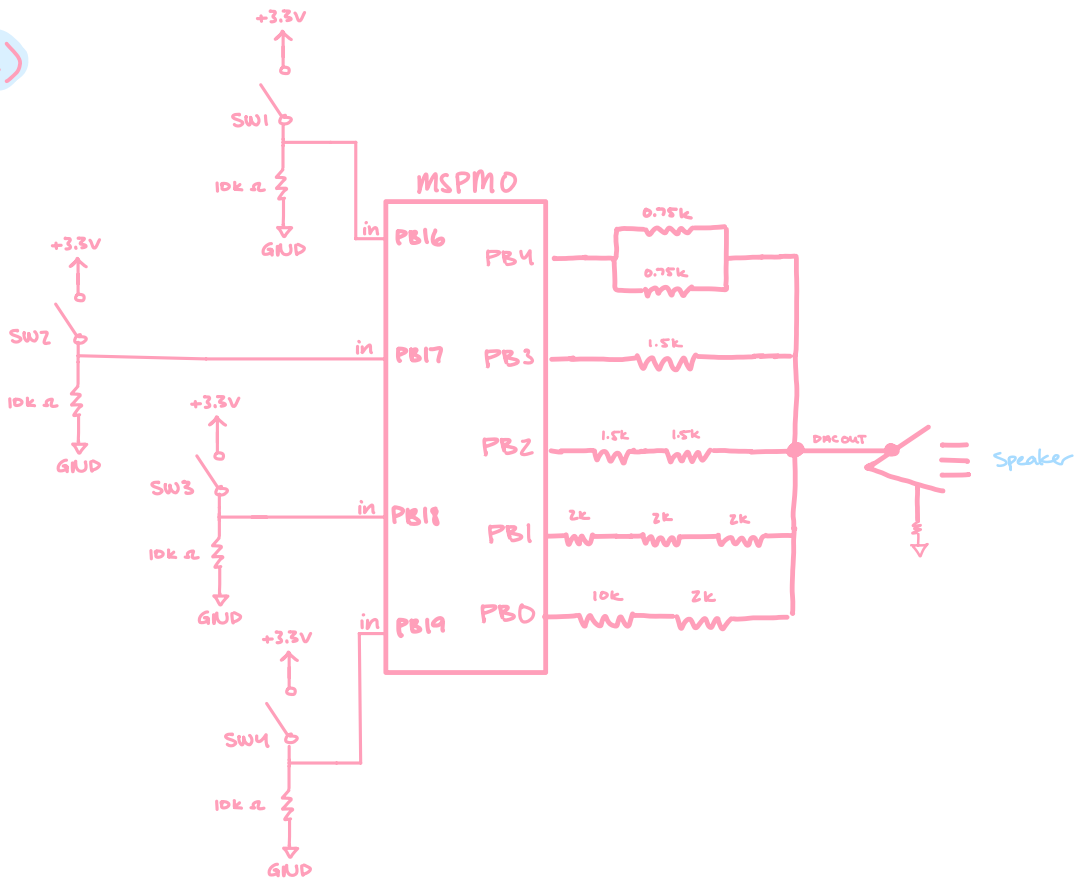
1) Melinda Dai

Dr. Holt

mmd3728

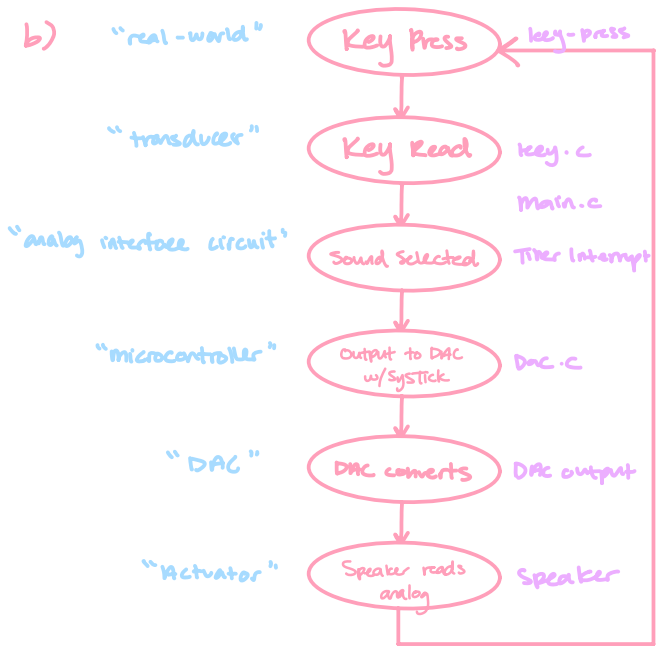
2)



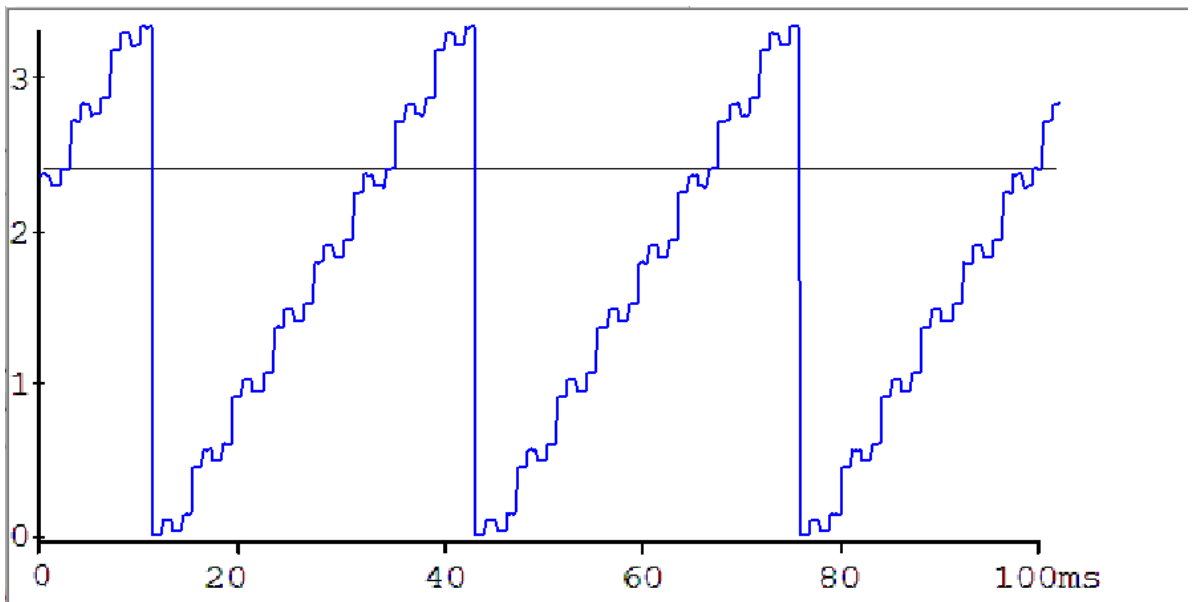
3) a)

SinTable[0] = 16	SinTable[17] = 13
SinTable[1] = 19	SinTable[18] = 10
SinTable[2] = 22	SinTable[19] = 8
SinTable[3] = 24	SinTable[20] = 5
SinTable[4] = 27	SinTable[21] = 4
SinTable[5] = 28	SinTable[22] = 2
SinTable[6] = 30	SinTable[23] = 1
SinTable[7] = 31	SinTable[24] = 1
SinTable[8] = 31	SinTable[25] = 1
SinTable[9] = 31	SinTable[26] = 2
SinTable[10] = 30	SinTable[27] = 4
SinTable[11] = 28	SinTable[28] = 5
SinTable[12] = 27	SinTable[29] = 8
SinTable[13] = 24	SinTable[30] = 10
SinTable[14] = 22	SinTable[31] = 13
SinTable[15] = 19	
SinTable[16] = 16	

3) b)



4)

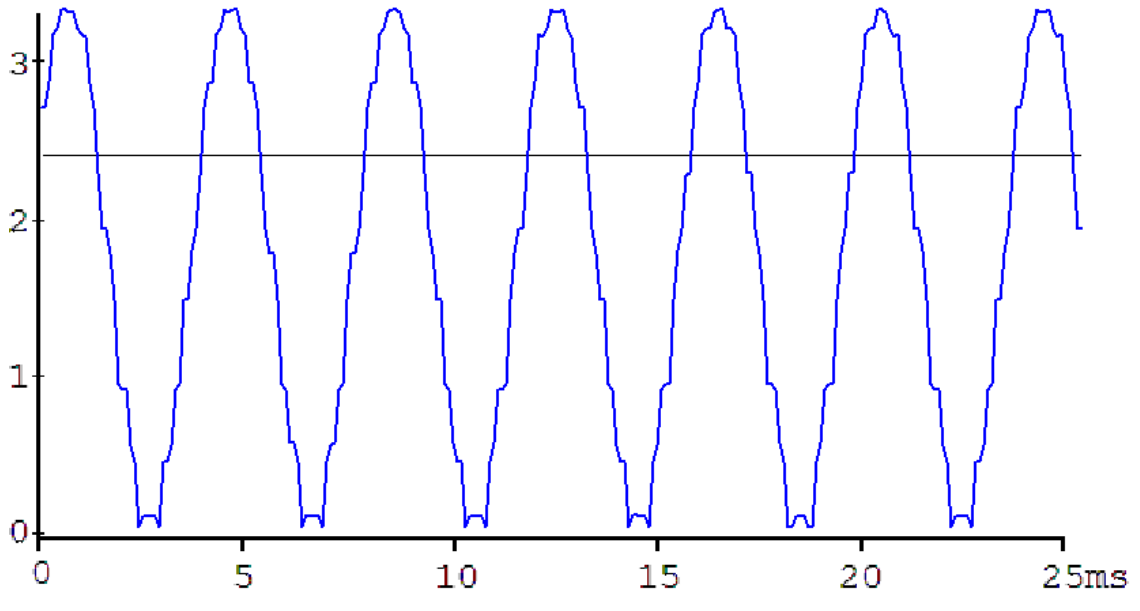


Ave=1.67V, Peak-peak=3.29V, Period=32.6ms, Freq= 31Hz
high-pulse=11.2ms, low-pulse=21.4ms

5)

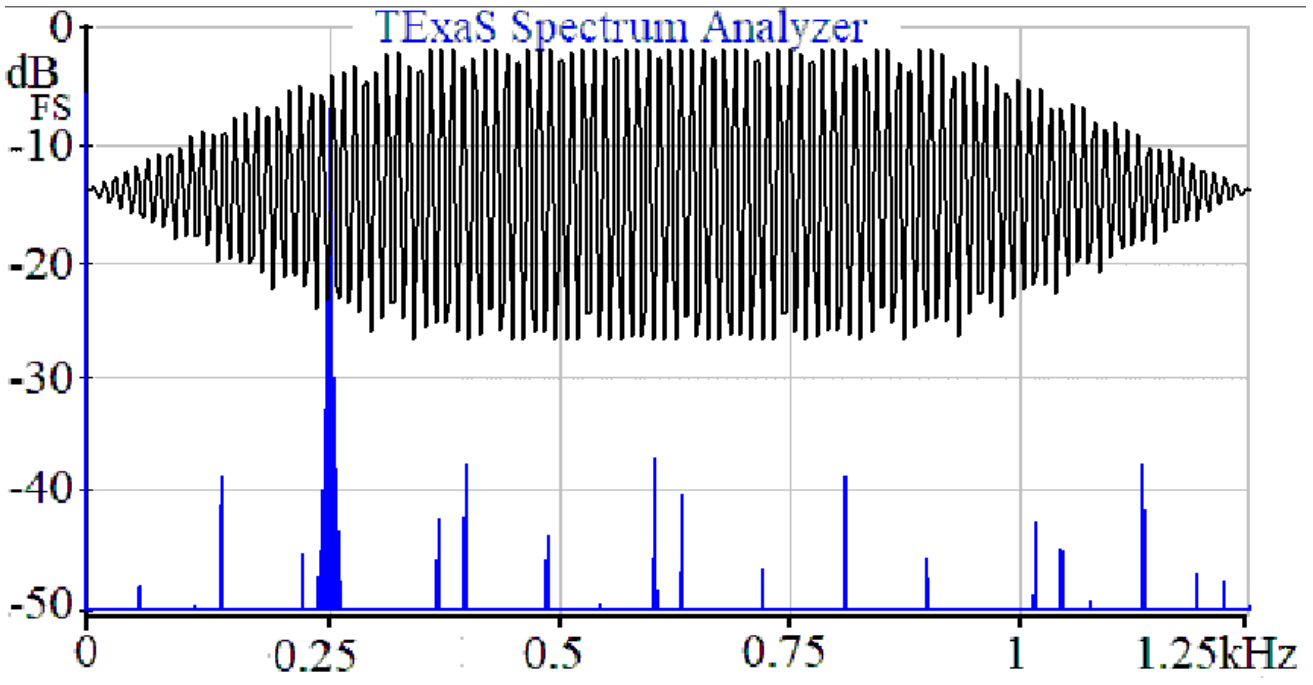
```
Lab 5, Spring 2026, Step 4. Debug switches
EID1= MMD3728
Switch= 0x1
Switch= 0x0
Switch= 0x2
Switch= 0x0
Switch= 0x4
Switch= 0x0
Switch= 0x8
Switch= 0x0
```

6)



Ave=1.77V, Peak-peak=3.26V, Period=4.0ms, Freq=251Hz
high-pulse=1.6ms, low-pulse=2.3ms

7) key 0:



fs=2500Hz, dF=2.44Hz, N=1024, Time=409.6ms
Ave voltage=1.69V, Max peak=261.2Hz, -7.2dB

8)

```
Lab 5, Spring 2026
Student EID1= MMD3728
Connect PB20 to DACOUT.
Initialization, good, Score=4
Activate piano Key0 Good. Score=10
Activate piano Key1 Good. Score=16
Activate piano Key2 Good. Score=22
Activate piano Key3 Good. Score=25
Done. Score=25
```

9) a)

Digital **Data**	Theoretical DAC voltage	Measured DAC voltage
0	0.00v	0.00v
1	0.106452v	0.105v
7	0.745161v	0.744v
8	0.851613v	0.849v
15	1.59677v	1.593v
16	1.70323v	1.712v
17	1.80968v	1.813v
23	2.44839v	2.446v
24	2.55484v	2.537v
25	2.66129v	2.672v
30	3.19355v	3.187v
31	3.3v	3.296v

9) b) Resolution = $\frac{\text{Range}}{\text{Levels}-1} = \frac{3.3}{2^5-1} = 0.103125v$

Range = Max V - Min V = 3.3 - 0 = 3.3v

Precision = 2^5 = 32 levels

Accuracy = $\frac{\text{Actual} - \text{Theoretical}}{\text{Theoretical}} = -0.19\%$ error

Digital **data**	Measured (V)	Theory (V)	|Error| (V)	|Error|/3.3V	Resolution (V)
0	0.000	0.000	0.000	0.00%	
1	0.105	0.106	0.001	0.04%	0.105
7	0.744	0.745	0.001	0.04%	
8	0.849	0.852	0.003	0.08%	0.105
15	1.593	1.597	0.004	0.11%	
16	1.712	1.703	0.009	0.27%	0.119
17	1.813	1.810	0.003	0.10%	0.101
23	2.446	2.448	0.002	0.07%	
24	2.537	2.555	0.018	0.54%	0.091
25	2.672	2.661	0.011	0.32%	0.135
30	3.187	3.194	0.007	0.20%	
31	3.296	3.300	0.004	0.12%	0.109
Average accuracy of full scale(V)= 0.005				0.16%	
Average resolution (V)= 0.109					

10) a) It occurs every time the systick canter counts down from 1 to 0

10) b) In the startup_mspm0g3507_ticlang.c

10) c)
1. Current instruction finished executing
2. Registers R0-R7, R12, LR, PC, & PSR pushed on stack
3. Vector address of systick_Handler loaded from vectorTable into PC
4. LR set to special value to indicate interrupt return
5. IPSR set to 15 (systick exception #)

10) d) When LR holds special exception return value, the processor recognizes it as an interrupt return and automatically pops R0-R3, R12, LR, PC, and PSR off the stack